

AIDS Survival Analysis

Blake Xu, Mentor: Rui Wang

Introduction & Data Description

Summary

AIDS is a serious infectious disease caused by human immunodeficiency virus (HIV). HIV primarily attacks the immune system by infecting and destroying CD4⁺ T cells, which play an important role in coordinating immune responses and protecting the body against infections. As CD4⁺ T-cell counts decline, patients become increasingly vulnerable to opportunistic infections and other AIDS-related complications.

In this report, we analyze data from a randomized clinical trial comparing two antiretroviral drugs, didanosine (ddI) and zalcitabine (ddC), in HIV-infected patients who had failed or were intolerant of zidovudine (AZT) therapy. The dataset is available from the JM package in R (JM package 2024). The survival outcome of interest is time to death, with patients who did not experience death during follow-up treated as censored observations. The key longitudinal biomarker is square-root transformed CD4 count, which was repeatedly measured over follow-up.

The main goal of this report is to examine how treatment group, baseline clinical variables, and longitudinal CD4 measurements are associated with mortality risk. We first use descriptive summaries and Kaplan-Meier curves to compare survival patterns across patient groups. We then fit a Cox proportional hazards model to estimate adjusted associations between baseline covariates and death, and we assess whether the proportional hazards assumption is reasonable.

Because CD4 count is repeatedly measured over time and reflects the patient's changing immune status, we then fit a joint model for longitudinal CD4 measurements and time to death. In the joint model, the longitudinal submodel describes the patient-specific CD4 trajectory, while the survival submodel links mortality risk to baseline covariates and the current underlying CD4 level.

Overall, the analysis suggests that CD4 is an important longitudinal biomarker for mortality risk. Lower current CD4 levels are associated with higher risk of death, and previous AIDS diagnosis or opportunistic infection is consistently associated with worse survival outcomes. The final joint model also suggests a possible treatment difference between ddI and ddC, although this treatment effect should be interpreted cautiously.

Descriptive Analysis

We will first take a look at what does our dataset look like.

Table 1: Preview of the original AIDS dataset.

patient	Time	death	CD4	obstime	drug	gender	prevOI	AZT	start	stop	event
1	16.97	0	10.677078	0	ddC	male	AIDS	intolerance	0	6.00	0
1	16.97	0	8.426150	6	ddC	male	AIDS	intolerance	6	12.00	0
1	16.97	0	9.433981	12	ddC	male	AIDS	intolerance	12	16.97	0
2	19.00	0	6.324555	0	ddI	male	noAIDS	intolerance	0	6.00	0
2	19.00	0	8.124038	6	ddI	male	noAIDS	intolerance	6	12.00	0
2	19.00	0	4.582576	12	ddI	male	noAIDS	intolerance	12	18.00	0
2	19.00	0	5.000000	18	ddI	male	noAIDS	intolerance	18	19.00	0
3	18.53	1	3.464102	0	ddI	female	AIDS	intolerance	0	2.00	0

The AIDS dataset is in long format, so each patient may appear in multiple rows because CD4 was measured repeatedly over time. The variable patient identifies each individual, and obstime records the time of each CD4 measurement. The variable CD4 is the square-root transformed CD4 count.

The survival outcome is described by Time and death. The variable Time gives the total follow-up time, and death is the patient-level event indicator, where death = 1 means death and death = 0 means censoring. The variables start and stop define each observation interval for CD4 count, while event indicates whether death occurred at the end of that interval.

The baseline covariates are drug, gender, prevOI, and AZT. The variable drug compares ddC with ddI, where ddC is zalcitabine and ddI is didanosine. The variable prevOI indicates whether the patient had a previous AIDS diagnosis. The variable AZT records prior AZT experience, where intolerance means the patient could not tolerate AZT and failure means AZT treatment was not effective.

For the descriptive table and baseline Cox model, the data were collapsed to one row per patient. The original long-format data were kept for CD4 trajectory visualization and later longitudinal or joint modeling.

Table 2: Characteristics by survival status

Characteristic	Censored N = 279	Death N = 188
Follow-up time	15.4 (2.8)	8.5 (4.6)
Gender		
female	26 (57.8%)	19 (42.2%)
male	253 (60.0%)	169 (40.0%)
Treatment group		
ddC	149 (62.9%)	88 (37.1%)
ddI	130 (56.5%)	100 (43.5%)
Previous opportunistic infection		
noAIDS	132 (82.5%)	28 (17.5%)
AIDS	147 (47.9%)	160 (52.1%)
AZT history		
intolerance	201 (68.8%)	91 (31.2%)
failure	78 (44.6%)	97 (55.4%)

[†] Mean (SD); n (%)

Table 2 summarizes patient characteristics by survival status. Overall, 279 patients were censored and 188 patients died during follow-up. The mean follow-up time was longer among censored patients than among patients who died, which is expected because deceased patients tend to have shorter observed survival times.

Several baseline characteristics show noticeable differences between the censored and death groups. Gender appears to differ slightly by survival status: among female patients, 57.8% were censored and 42.2% died, while among male patients, 60.0% were censored and 40.0% died. This difference is relatively small. Treatment group also shows some difference: patients receiving ddC had a higher censored proportion than those receiving ddI, suggesting better observed survival in the ddC group in this sample.

The most striking difference is previous opportunistic infection status. Among patients with no previous AIDS diagnosis, 82.5% were censored and only 17.5% died. In contrast, among patients with previous AIDS, only 47.9% were censored and 52.1% died. This suggests that previous opportunistic infection is strongly associated with worse survival outcomes. AZT history also appears related to survival: patients with AZT intolerance had a higher censored proportion, while patients with AZT failure had a higher death proportion.

These descriptive results suggest that previous opportunistic infection, AZT history, and possibly treatment group may be important predictors of survival. However, these comparisons are unadjusted, so formal survival analysis, such as Kaplan-Meier curves, log-rank tests, and Cox proportional hazards models, is needed to assess these relationships more rigorously.

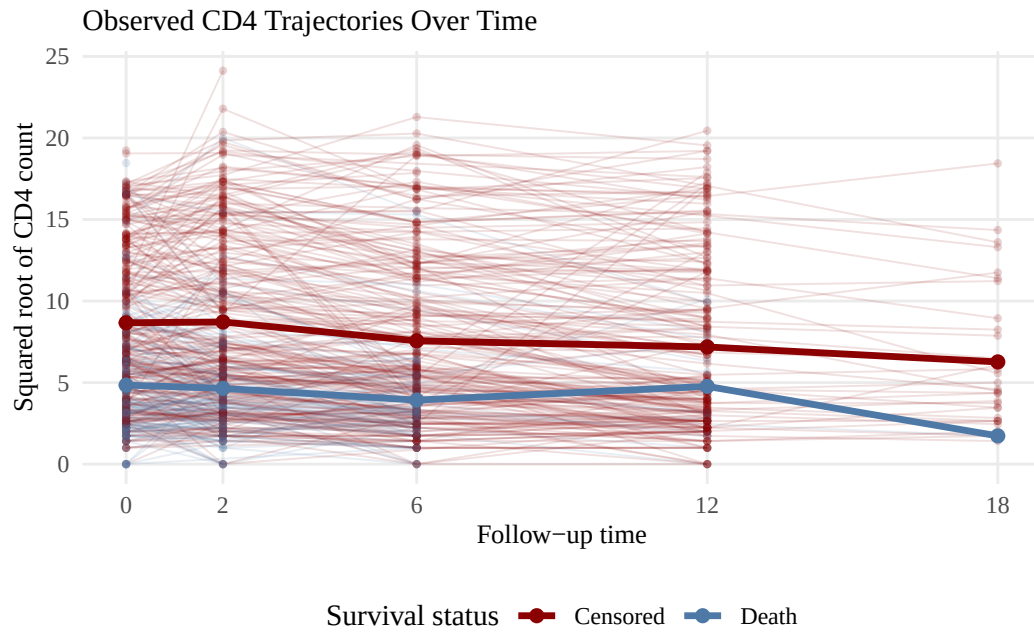


Figure 1: Observed CD4 trajectories by survival status.

As shown in Figure 1, patients who died generally had lower observed CD4 counts throughout follow-up compared with censored patients. The average CD4 trajectory for the death group remained below that of the censored group at each recorded time point. Although individual CD4 measurements fluctuate over time, the overall difference between the two groups is relatively clear. This suggests that lower CD4 levels may be associated with higher mortality risk.

Kaplan-Meier Analysis

To compare survival experience across patient groups, Kaplan-Meier survival curves were estimated. In survival analysis, the outcome of interest is the time until an event occurs. In this study, the event is death. Let (T) denote a patient's survival time. The survival function is defined as

$$S(t) = P(T > t),$$

which represents the probability that an individual survives beyond time t .

A key feature of survival data is censoring. In this dataset, some patients did not die during the observed follow-up period. For these patients, the exact survival time is unknown; we only know that they survived at least until their last observed follow-up time. These observations are right-censored. Standard methods that ignore censoring may produce biased estimates of survival, so a method that accounts for censoring is needed.

The Kaplan-Meier estimator is a nonparametric estimator of the survival function that accounts for right-censored observations.

Suppose $(t_1 < t_2 < \dots < t_k)$ are the distinct observed event times. At each event time (t_j) , let (d_j) be the number of events that occur at (t_j) , and let (n_j) be the number of individuals at risk immediately before (t_j) . The Kaplan-Meier estimator is

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

At each observed event time, the quantity

$$1 - \frac{d_j}{n_j}$$

represents the estimated conditional probability of surviving beyond (t_j) , given that the individual has survived up to (t_j) . Therefore, the Kaplan-Meier estimator estimates overall survival beyond time (t) by multiplying these conditional survival probabilities over all event times up to (t) .

Censored individuals are included in the risk set until their censoring time, but they do not count as events. After censoring, they are no longer included in later risk sets. In this way, the Kaplan-Meier estimator uses all available follow-up information while properly accounting for incomplete survival times.

Kaplan-Meier curves were used to visually compare survival patterns across treatment group, gender, previous opportunistic infection status, and AZT history. Larger separation between curves suggests greater differences in survival experience between groups.

To formally compare survival curves between 2 groups, the log-rank test was used. The null hypothesis is that the survival functions are equal across groups:

$$H_0 : S_1(t) = S_2(t) \quad \forall t.$$

The log-rank test compares the number of observed events in one group with the number of expected events under the null hypothesis. For two groups, the test statistic can be written as

$$X^2 = \frac{(O_1 - E_1)^2}{V_1} \sim \chi_1^2$$

where

(O_1) = total observed number of events for group 1.

(E_1) = total expected number of events in group 1 under the null hypothesis.

(V_1) = total variance of $(O_1 - E_1)$.

Under the null hypothesis, this statistic approximately follows a chi-square distribution with one degree of freedom.

Kaplan-Meier curves were plotted for treatment group, gender, previous opportunistic infection status, and AZT history. The corresponding p-values shown in Figure 2 are from log-rank tests.

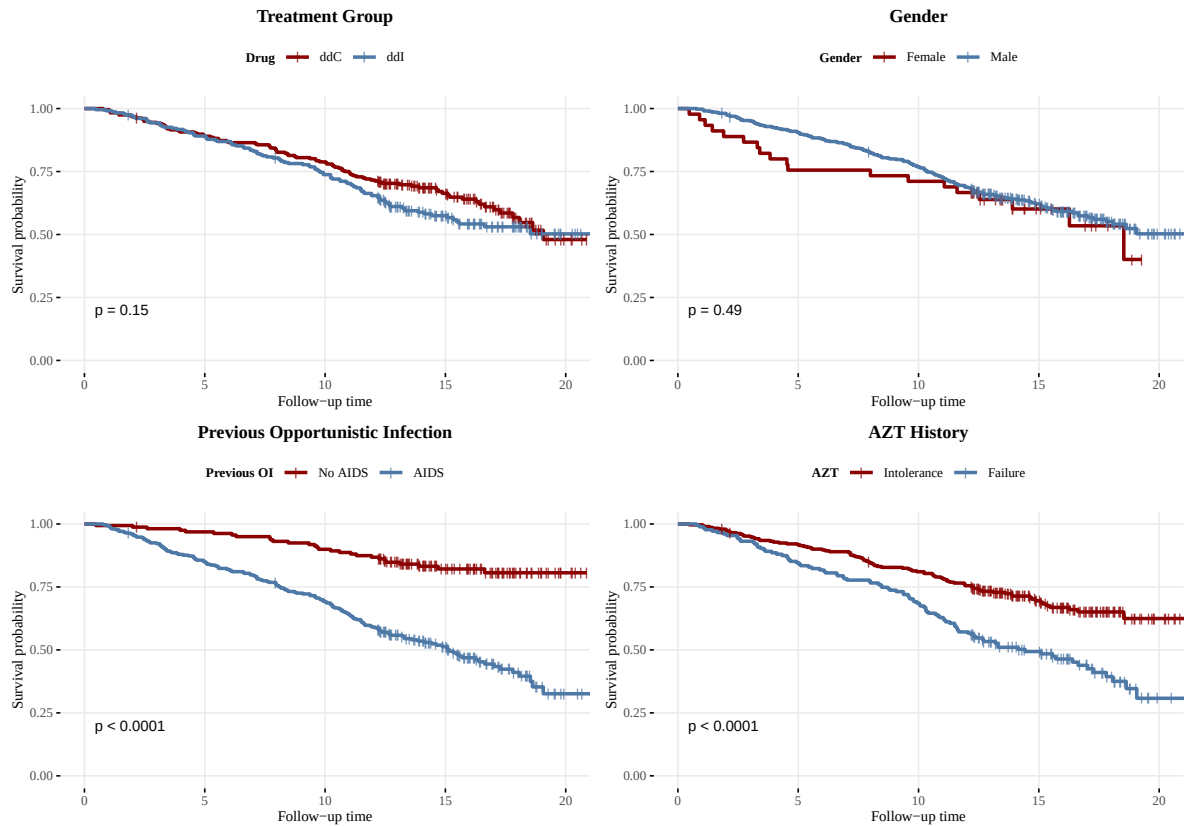


Figure 2: Kaplan-Meier survival curves stratified by baseline covariates.

Figure 2 presents Kaplan-Meier survival curves stratified by treatment group, gender, previous opportunistic infection status, and AZT history. The survival curves for previous opportunistic infection and AZT history show clear separation over time. In particular, patients **with** previous AIDS diagnosis have substantially **lower** survival probabilities than patients with no previous AIDS diagnosis. Similarly, patients **with** AZT failure have **lower** survival probabilities than patients with AZT intolerance. The log-rank test p-values for both variables are below 0.05, suggesting statistically significant differences in survival distributions across these groups. These findings are consistent with the descriptive results in Table 1, where previous opportunistic infection and AZT failure were associated with higher proportions of death.

In contrast, the survival curves for treatment group and gender show much weaker separation. The log-rank p-values for treatment group and gender are 0.15 and 0.49, respectively, so there is not strong evidence of a difference in survival between drug groups or gender groups based on the unadjusted Kaplan-Meier analysis.

The shapes of some curves also suggest that the proportional hazards assumption should be checked formally. For example, the treatment-group curves appear close together for much of the follow-up

period and may change separation over time. The gender curves also show changing separation, with female and male survival probabilities becoming closer later in follow-up. However, Kaplan-Meier curves alone cannot confirm whether the proportional hazards assumption holds. Therefore, formal diagnostic tools, such as Schoenfeld residual tests, are needed before relying on a Cox proportional hazards model.

These unadjusted Kaplan-Meier results motivate fitting a Cox proportional hazards model to estimate the association between each covariate and survival while adjusting for the other baseline variables.

Cox Proportional Hazards

The Kaplan-Meier curves and log-rank tests provide unadjusted comparison of survival across groups. However, these methods only examine one covariate at a time and do not adjust for other patient characteristics. To examine the association between baseline covariates and mortality rate while controlling for other covariates, a Cox proportional hazard model was fitted.

$$h_i(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T_i < t + \Delta t \mid T_i \geq t)}{\Delta t},$$

which represents the instantaneous risk of death at time (t) , given that the patient has survived up to time (t) .

The Cox proportional hazard model can be written as

$$h_i(t \mid X_i) = h_0(t) \exp(\beta^T X_i),$$

Where

h_i = the hazard of a patient (i) .

h_0 = the unspecified baseline hazard function.

X_i = a vector of covariates of patient (i) .

β = a vector of regression coefficients.

PH diagnostics

The Cox proportional hazards model assumes that, after controlling for other covariates, the ratio of the hazards between two individuals remains constant over time. In other words, after controlling for the other variables in the model, the relative hazard comparing two groups should not systematically change as follow-up time increases. If this assumption is violated, then a single constant hazard ratio may not adequately describe the covariate effect.

To illustrate, for two patients with covariate vectors (X_i) and (X_k) , the hazard ratio is

$$\frac{h(t|X_i)}{h(t|X_k)} = \exp \beta^T (X_i - X_k).$$

Because the baseline hazard $(h_0(t))$ cancels out, this hazard ratio does not depend on time (t) . In other words, the Cox model allows the baseline hazard to vary over time, but it assumes that the relative effect of a covariate is constant over time. This assumption will be assessed using Schoenfeld residual diagnostics.

Schoenfeld residuals are calculated at observed event times and can be used to examine whether covariate effects vary over time. In R, this diagnostic was performed using the `cox.zph()` function, which tests whether the scaled Schoenfeld residuals are associated with time. A non-significant p-value suggests no strong evidence against the proportional hazards assumption, while a significant p-value suggests that the effect of that covariate may change over time.

The diagnostic test was applied to the Cox model:

$$h_i(t | X_i) = h_0(t) \exp (\beta_0 + \beta_1 \text{drug}_i + \beta_2 \text{gender}_i + \beta_3 \text{prevOI}_i + \beta_4 \text{AZT}_i).$$

For each covariate, the null hypothesis is that the corresponding coefficient is constant over time:

$$H_0 : \beta_j(t) = \beta_j.$$

Table 3 reports the Schoenfeld residual test results for each covariate and for the global model. Large p-values indicate no strong evidence against the proportional hazards assumption.

Table 3: Schoenfeld residual test for the proportional hazards assumption.

Covariate	chisq	df	p
gender	2.499	1	0.114
AZT	0.470	1	0.493
GLOBAL	2.873	4	0.579
drug	0.120	1	0.730
prevOI	0.084	1	0.773

As shown in Table 3, all covariate-specific p-values are greater than 0.05, and the global test is also not significant. Therefore, there is no strong evidence against the proportional hazards assumption for this Cox model.

Baseline Cox Model

Therefore, we fit a baseline Cox proportional hazards model including treatment group, gender, previous opportunistic infection status, and AZT history as baseline covariates. The model is

$$h_i(t|X_i) = h_0(t) \exp(\beta_0 + \beta_1 \text{drug}_i + \beta_2 \text{gender}_i + \beta_3 \text{prevOI}_i + \beta_4 \text{AZT}_i)$$

where $h_i(t | X_i)$ is the hazard for patient i , h_0 is the baseline hazard, and the exponentiated coefficients are interpreted as adjusted hazard ratios. Each hazard ratio represents the association between one covariate and risk while holding the other covariates constant.

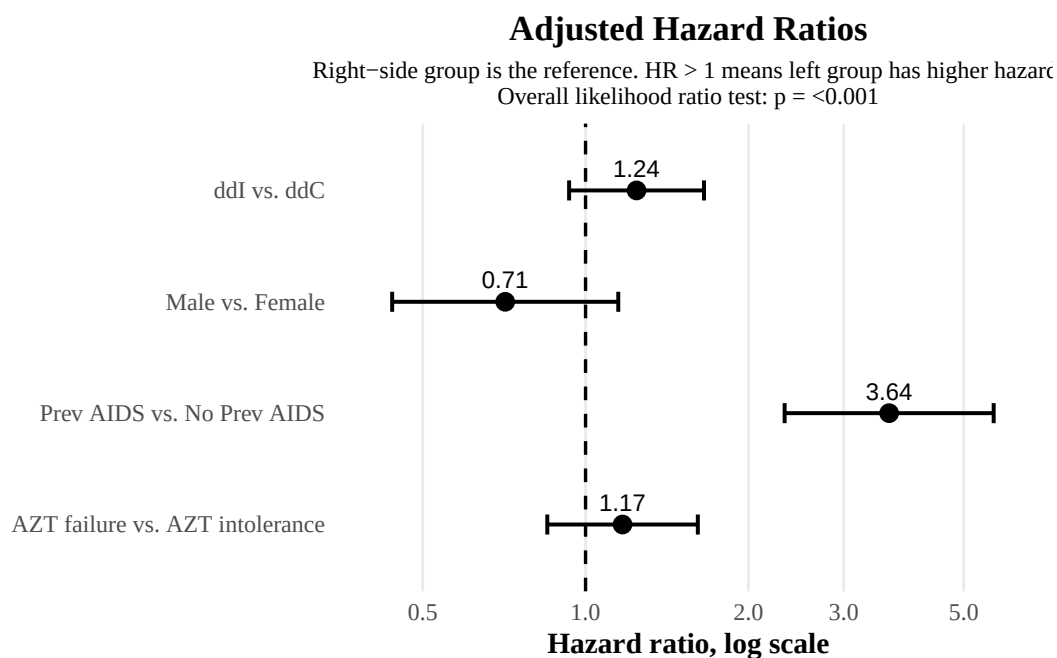


Figure 3: Adjusted hazard ratios from the baseline Cox PH model.

Figure 3 presents the adjusted hazard ratios from the baseline Cox proportional hazards model. The most notable result is previous opportunistic infection status. Patients with previous AIDS diagnosis had an estimated hazard ratio of 3.64 compared with patients with no previous AIDS diagnosis, holding treatment group, gender, and AZT history constant. This means that the estimated death hazard

for patients with previous AIDS diagnosis was about 3.64 times the hazard for patients without previous AIDS diagnosis. The confidence interval does not include 1, indicating that this association is statistically significant.

The estimated hazard ratio for males compared with females was 0.71, suggesting that males had a lower estimated death hazard than females after adjustment. However, the confidence interval includes 1, so this difference is not statistically significant. Similarly, patients receiving didanosine had an estimated hazard ratio of 1.24 compared with patients receiving zalcitabine, and patients with AZT failure had an estimated hazard ratio of 1.17 compared with patients with AZT intolerance. Both confidence intervals include 1, suggesting that these covariates are not statistically significant predictors of mortality in the adjusted Cox model.

Overall, the Cox model supports the findings from the Kaplan-Meier analysis: previous opportunistic infection status is the strongest baseline predictor of mortality in this dataset. Although treatment group, gender, and AZT history show some differences in point estimates, their confidence intervals include 1, so there is not strong evidence that they are independently associated with death hazard after adjustment.

Joint Modeling

In the baseline survival analysis, we modeled the time-to-death outcome using baseline covariates such as treatment group, gender, previous opportunistic infection, and AZT history. However, CD4 count is an important biomarker of immune function and is strongly related to disease progression in HIV/AIDS studies. Therefore, instead of only using baseline covariates, we would like to incorporate the longitudinal CD4 information into the survival model to study whether changes in CD4 over follow-up are associated with mortality risk.

A natural first approach would be to include CD4 count directly as a time-varying covariate in a Cox proportional hazards model. However, this approach has several limitations. First, CD4 count is an endogenous time-dependent biomarker, meaning that it is part of the disease process itself and is related to the patient's underlying health status. Patients with worse disease progression may tend to have lower CD4 counts and higher risk of death. Therefore, CD4 is not an external covariate like gender or treatment assignment. Second, observed CD4 measurements are subject to measurement error, so the recorded CD4 value may not exactly represent the patient's true underlying immune status. Third, CD4 is only observed at scheduled visit times rather than continuously over follow-up. If CD4 is directly used as a time-varying covariate, the usual approach often carries the last observed value forward between two visits, implicitly assuming that CD4 remains constant between (t_1) and (t_2) . This may be unrealistic because CD4 evolves dynamically over time.

To address these limitations, we treat the observed CD4 measurements as noisy observations of an underlying, subject-specific CD4 process that evolves continuously over time. Rather than using the recorded CD4 values as a patient's true immune status, we model each individual's latent CD4 trajectory and examine how it relates to the risk of death. This motivates the use of a joint model for

longitudinal and survival data, which simultaneously models repeated CD4 measurements and time to death through shared subject-specific information. In doing so, it accounts for measurement error, irregular observation times, within-subject correlation, and the endogenous relationship between CD4 evolution and mortality risk.

We begin with the standard current value association structure. Let $y_i(t)$ denote the observed CD4 count for subject (i) at time (t). The longitudinal submodel is written as

$$y_i(t) = m_i(t) + \varepsilon_i(t),$$

where

m_i = True CD4 count trajectory for subject (i).

ε_i = Measurement error.

The true CD4 trajectory is modeled using a mixed-effects model,

$$m_i(t) = x_i^\top(t)\beta + z_i^\top(t)b_i,$$

where

x_i = fixed-effects covariates effect describing the population-average CD4 trajectory.

β = corresponding fixed-effect coefficient vector.

z_i = covariates for the random-effects structure.

b_i = subject-specific random effect.

Usually, we assume

$$b_i \sim N(0, D), \quad \varepsilon_i(t) \sim N(0, \sigma^2).$$

The survival submodel links this underlying CD4 trajectory to the hazard of death. Under the current value association structure, the hazard for subject (i) at time (t) is

$$h_i(t) = h_0(t) \exp(\gamma^\top w_i + \alpha m_i(t))$$

h_0 = baseline hazard function.

w_i = A vector of baseline covariates.

γ = corresponding regression coefficients.

α = association between current CD4 level and the risk of death.

The parameter α is the key parameter connecting the CD4 trajectory and survival processes. To illustrate, if $\alpha < 0$, then higher CD4 levels are associated with lower hazard(risk of death). If $\alpha > 0$, then higher CD4 levels are associated with higher hazard. Therefore, the current value joint model directly addresses whether a patient's current immune status, represented by the latent trajectory m_i , is associated with mortality risk

Modeling the Baseline Hazard Function

After introducing the current value joint model, we next specify how the baseline hazard function $h_0(t)$ is modeled. In a standard Cox proportional hazards model, the baseline hazard is often left unspecified because the regression coefficients can be estimated through the partial likelihood.

However, as discussed by Rizopoulos (2012), leaving the baseline hazard completely unspecified in a joint model may lead to underestimated standard errors. To avoid this issue, the baseline hazard can be modeled using a flexible parametric form. One common approach is to approximate the logarithm of the baseline hazard using spline basis functions Rizopoulos (2012). This allows the baseline risk of death to vary smoothly over time.

In this report, we model the logarithm of the baseline hazard using cubic splines. Specifically, we write

$$\log h_0(t) = \sum_{q=1}^Q \eta_q B_q(t, g),$$

where

$B_q = q_{th}$ cubic spline basis function.

η_q = corresponding coefficient of basis function.

Q = number of knots + 4.

g = degree of basis function.

Equivalently, the baseline hazard function can be written as the following

$$h_0(t) = \exp \left(\sum_{q=1}^Q \eta_q B_q(t) \right).$$

We will select the number of knots by comparing candidate models, with 1, 2, and 3 knots.

Table 4: Comparison of Baseline Hazard Spline Specifications

Baseline Hazard	Log-Likelihood	AIC	BIC
cubic B-spline, 1 internal knot	-4333.760	8699.520	8765.861
cubic B-spline, 2 internal knots	-4333.492	8700.983	8771.471
cubic B-spline, 3 internal knots	-4332.096	8700.191	8774.825

Table 4 compares cubic spline for the log baseline hazard with 1, 2, and 3 knots. Both AIC and BIC selected the model with 1 knot. Since the differences in AIC were small and the 1-knot model was the most parsimonious, we used the cubic spline with 1 knot to model the baseline hazard function.

Choosing Mixed-Effects Model

Before fitting the joint model, we first examine the longitudinal trajectory of CD4 over time. Inspired by the idea used by Mchunu et al. (2022), who plotted randomly selected individual CD4 profiles to assess whether the CD4 trajectory appeared linear or nonlinear, we randomly select 12 patients and plot their individual CD4 trajectories over time. This visual check helps motivate whether a simple linear mixed-effects model is sufficient or whether a more flexible spline-based mixed-effects model should be considered. If many patients show curved or irregular patterns, then a spline-based mixed-effects model may be more appropriate than a simple linear time model.

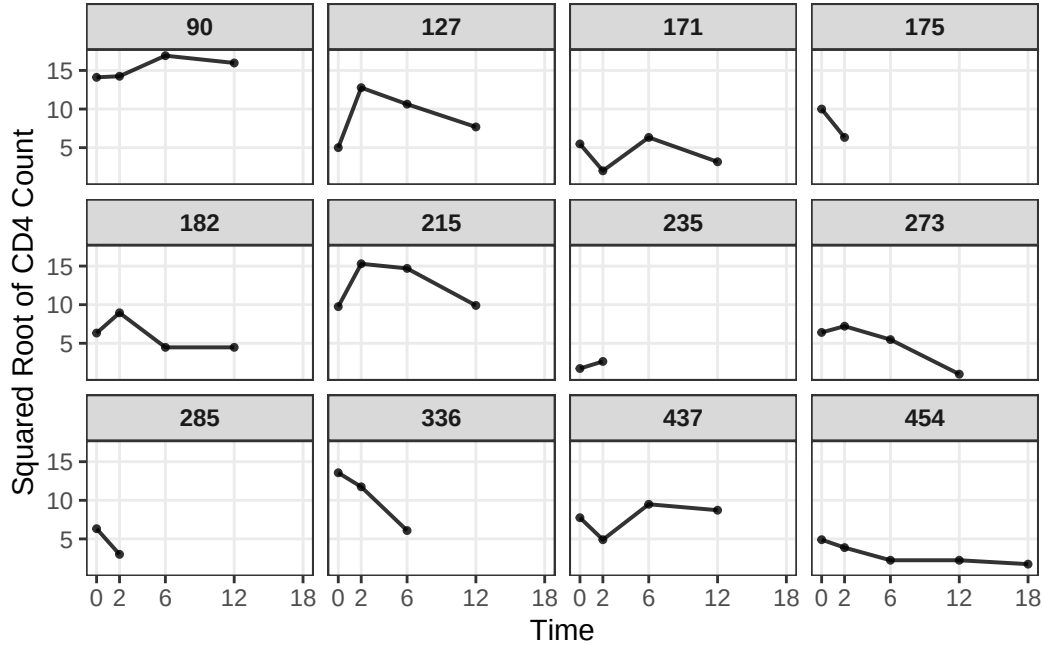


Figure 4: Observe Individual Trajectories of CD4 Count

Although some individual trajectories appear to show nonlinear patterns, each patient has only a small number of observed CD4 measurements, with at most four repeated observations. Therefore, the observed profiles are not sufficient to strongly conclude that the underlying square root CD4 trajectory is nonlinear.

Thus, we will fit and compare several candidate mixed-effects models for the longitudinal submodel. These candidate models are compared using AIC, BIC, and deviance. Specifically, we consider a linear mixed-effects model and natural cubic spline models with 1, 2 knots. These candidate models are compared using AIC, BIC, and deviance. We only consider 2 knots because we only have 4 observe time points.

Table 5: Comparison of Candidate Longitude Mixed-Effects Models

Model	AIC	BIC	Deviance
Linear	7133.522	7165.009	7121.522
Natural cubic spline: 1 knot	7135.158	7171.892	7121.158
Natural cubic spline: 2 knots	7131.545	7173.527	7115.545

Table 4 compares the candidate longitudinal mixed-effects models using AIC, BIC, and deviance. The natural cubic spline model with two internal knots has the smallest AIC and deviance, while the linear time model has the smallest BIC. However, the differences in AIC and deviance are small, suggesting that the more flexible spline model does not substantially improve model fit. In addition, because each patient has only a small number of observed CD4 measurements, a highly flexible spline trajectory may not be well supported by the data. Therefore, we select the linear mixed-effects model with a linear effect of time as the longitudinal submodel for the joint model.

Therefore, we choose the linear mixed-effects model as the longitudinal submodel. In addition, we model the log baseline hazard using a cubic B-spline with 1 internal knot. The final joint model is specified as follows.

Let Y_{ij} denote the observed square root CD4 count for patient i at observation time t_{ij} . The longitudinal submodel is

$$Y_{ij} = m_i(t_{ij}) + \varepsilon_{ij},$$

where the true underlying square root CD4 trajectory is modeled as

$$m_i(t_{ij}) = \beta_0 + \beta_1 t_{ij} + b_{0i} + b_{1i} t_{ij}.$$

Here, β_0 and β_1 represent the population-level intercept and time slope, while b_{0i} and b_{1i} represent the patient-specific random intercept and random slope. We assume

$$\varepsilon_{ij} \sim N(0, \sigma^2), \quad (b_{0i} \ b_{1i}) \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, D \right].$$

The survival submodel links the hazard of death to both baseline covariates and the current value of the underlying CD4 trajectory:

$$h_i(t \mid w_i, m_i(t)) = h_0(t) \exp(\gamma^\top w_i + \alpha m_i(t)),$$

where

$$w_i = (\text{drug}_i, \text{gender}_i, \text{prevOI}_i, \text{AZT}_i)^\top.$$

The baseline hazard is modeled flexibly using a cubic B-spline with 1 internal knot:

$$h_0(t) = \exp \left(\sum_{q=1}^5 \eta_q B_q(t, 3) \right).$$

Here, $B_q(t)$ is the q th cubic B-spline basis function for time, and η_q is its corresponding coefficient.

Fitting Final Joint Model

We will now fit our final model as shown above in R.

Table 6: Survival Submodel Estimates from the Final Joint Model

Parameter	Value	Std.Err	z-value	p-value	HR	Lower95	Upper95
drugddI	0.332	0.153	2.169	0.03	1.393	1.032	1.880
gendermale	-0.345	0.258	-1.337	0.181	0.708	0.427	1.174
prevOIAIDS	0.621	0.240	2.585	0.01	1.861	1.162	2.982
AZTfailure	0.132	0.169	0.785	0.433	1.141	0.820	1.588
Assoct	-0.255	0.038	-6.674	<0.001	0.775	0.719	0.835
bs1	-3.403	0.496	-6.861	<0.001	0.033	0.013	0.088
bs2	-2.560	0.609	-4.201	<0.001	0.077	0.023	0.255
bs3	-2.477	0.783	-3.163	0.002	0.084	0.018	0.390
bs4	-2.399	0.747	-3.212	0.001	0.091	0.021	0.393
bs5	-3.729	1.889	-1.974	0.048	0.024	0.001	0.973

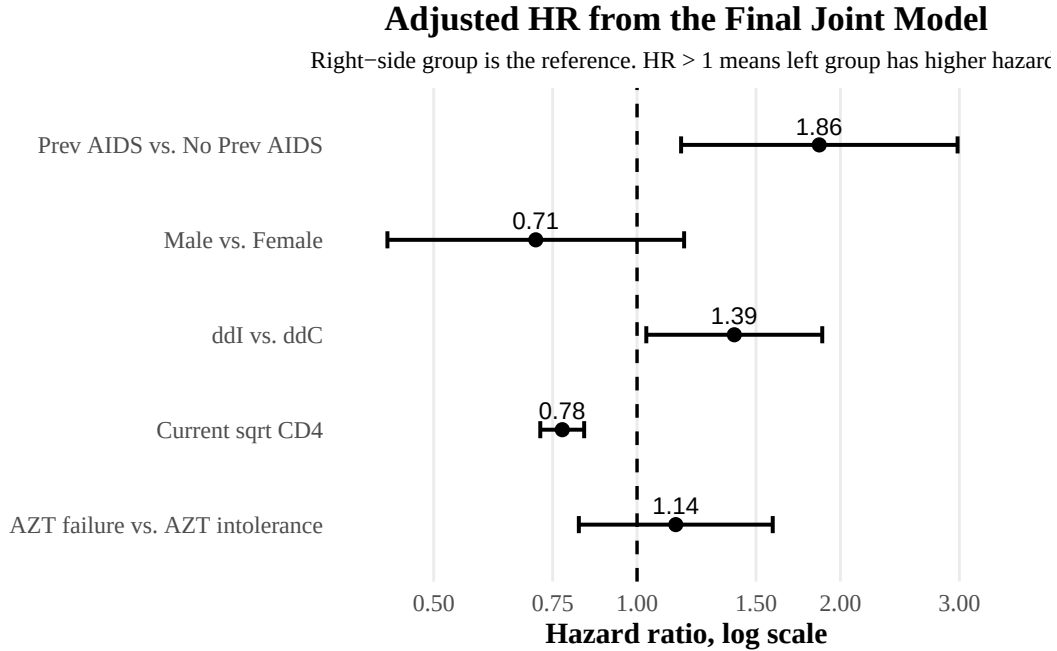


Figure 5: Adjusted hazard ratios from the final joint model.

The final joint model shows that current sqrt CD4 is strongly associated with the hazard of death. The association parameter is negative, with $\alpha = -0.255$, corresponding to a hazard ratio of $HR = 0.775$. This means that, holding other covariates constant, a one-unit increase in current sqrt CD4 is associated with approximately a 22.5% reduction in the hazard of death. This effect is statistically significant with $p < 0.001$.

After adjusting for current sqrt CD4 and other baseline covariates, treatment with ddI, didanosine, is associated with a higher hazard of death compared with ddC, zalcitabine. The estimated hazard ratio is $HR = 1.393$, meaning that patients in the ddI group have about 1.39 times the hazard of death compared with patients in the ddC group. This effect is statistically significant with $p = 0.03$.

Previous AIDS diagnosis or opportunistic infection remains an important risk factor. Patients with previous AIDS diagnosis have about 1.86 times the hazard of death compared with those without previous AIDS diagnosis, with $HR = 1.861$ and $p = 0.01$. This result is consistent with the baseline survival model.

In contrast, gender and AZT status are not statistically significant in the final joint model. The estimated hazard ratios are $HR = 0.708$ for males compared with females and $HR = 1.141$ for AZT failure compared with AZT intolerance, but both confidence intervals include 1.

Overall, the final joint model suggests that lower current CD4 level is strongly associated with increased mortality risk. In addition, previous AIDS diagnosis and treatment with didanosine are associated with

higher hazard of death, while gender and AZT status do not show clear evidence of association after adjustment.

Discussion & Limitations

The final joint model suggests that CD4 is a highly informative longitudinal biomarker for mortality risk in patients with HIV/AIDS. This finding is clinically reasonable because CD4 reflects immune system function and HIV disease progression. By incorporating repeated CD4 measurements, the joint model provides a more complete picture than the baseline Cox model, which only uses baseline covariates.

Previous AIDS diagnosis or opportunistic infection also remains an important risk factor across the analysis. This result is consistent with the descriptive summaries, Kaplan-Meier curves, baseline Cox model, and final joint model. Therefore, previous opportunistic infection appears to be a stable predictor of worse survival outcomes in this dataset.

The treatment effect should be interpreted with caution. Although the final joint model suggests that didanosine may be associated with higher mortality risk than zalcitabine, the evidence is not extremely strong. The p-value is close to 0.05, so this result should not be overinterpreted as definitive evidence that the two treatments have meaningfully different effects on survival. Instead, it suggests a possible treatment difference that would require further investigation.

One important limitation is the limited number of CD4 observation times. CD4 was only measured at a few scheduled visits, such as baseline, 2, 6, 12, and 18 months. Because many patients have only a small number of repeated measurements, the estimated individual CD4 trajectories may not fully capture the true underlying CD4 process. This could make the estimated association between current CD4 and mortality less precise.

Another limitation is the choice of a linear mixed-effects model for the longitudinal CD4 process. This choice is reasonable because the CD4 measurements are sparse, but it may still be too simple for patients whose CD4 trajectories change nonlinearly over time. More flexible longitudinal models could be considered, but they may not be well supported by the limited number of observations per patient.

Moreover, this report only considered the current value association structure in the final joint model. Under this structure, the hazard of death is linked to the patient's current underlying sqrt CD4 level. However, other association structures may also be clinically meaningful. For example, a cumulative effects structure links the hazard of death to the accumulated CD4 history over time, rather than only the current CD4 value. (Mchunu et al. 2022) mention that the cumulative effects parameterization has been shown to increase the statistical power of the analysis.

In our dataset, CD4 was measured at only a few scheduled observation times, such as baseline, 2, 6, 12, and 18 months. Because the longitudinal CD4 information is sparse, estimating the full cumulative CD4 history may be unstable or imprecise. Therefore, the current value structure was used as a simpler and more interpretable approach. Future work could compare current value, cumulative effects, weighted

cumulative effects, and slope-based association structures to assess whether the main conclusions are sensitive to the way CD4 is linked to mortality risk.

Finally, the results should be interpreted as model-based associations rather than definitive causal conclusions. Although the data come from a clinical trial, the joint model estimates still depend on assumptions about the longitudinal process, the survival process, censoring, and the baseline hazard specification.

Overall, the analysis shows that incorporating longitudinal CD4 measurements provides important information about mortality risk. The joint model supports the conclusion that CD4 is strongly related to survival, while also suggesting that previous opportunistic infection is an important baseline risk factor. The possible treatment difference between didanosine and zalcitabine should be viewed cautiously and may require further study.

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